Digital Curation and Datafication of Byzantine Seal Monograms

From Annotation to Classification

Ezéchiel Sawadogo Mamadou D. Barry Saranga K. Mahanta Victoria Eyharabide

Sorbonne Université

28th July, 2026

Byzantine Lead Seals

- Byzantine lead seals are a major source for studying the Byzantine Empire.
- They identify their owners through Greek inscriptions, iconographic scenes, and **monograms**.
- Monograms can lead to multiple valid readings.
- Degradations add on to an already complex problem.



A block monogram. The same arrangement of strokes can lead to two different readings: Ἀναστασίου (Anastasiou) (genitive of Anastasios) or Ἰουστινιανοῦ (Ioustinianou) (genitive of Ioustinianos).

Why seals need curation-aware AI?

- Monograms are valuable but difficult to process at scale because image quality varies strongly across objects and collections.
- Digital access therefore requires more than digitisation: it needs metadata about legibility, uncertainty, and computational reliability.

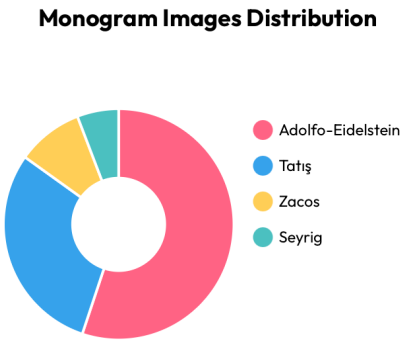
The Heritage Problem

- A seal photograph is not a neutral input: corrosion, wear, lighting, and relief can hide or distort the monogram.
- Downstream tools can produce confident-looking outputs even when the object is barely readable.
- **Which images are reliable enough for which computational tasks?**

Dataset and Quality Level Annotations

Collection	Total	M0	M1	M2	M3
Adolfo-Eidelstein	**350**	48	159	125	18
Zacos, Tatış, Seyrig	**275**	32	113	119	11
Total	**625**	80	272	244	29

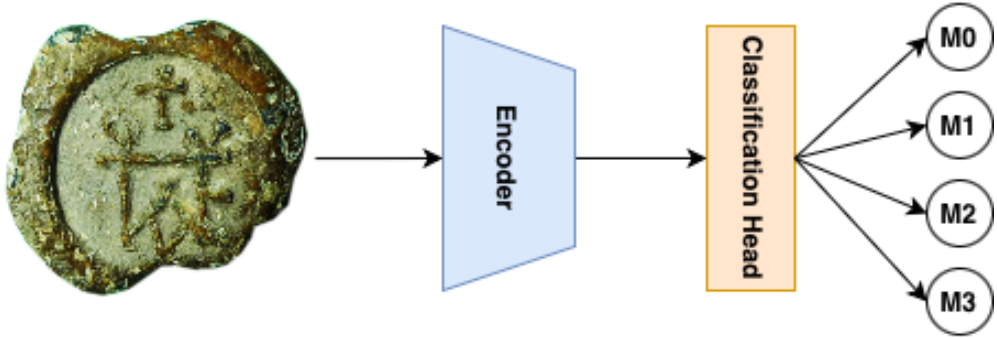
Class	Interpretation
M0	Excellent condition; good contrast.
M1	Good condition; minor defects or reduced contrast.
M2	Degraded seal; broken components; partial legibility.
M3	Heavily degraded; monogram barely readable.



Dataset and Quality Level Annotations

Collection	M0	M1	M2	M3
Adolfo	 101__5.76-77__335	 102__5.78__169	 100__5.75__215	 7__2.2__484
Tatish	 Tatish 1748_B	 Tatish 1748_A	 Tatish 1769_B	 Tatish 2362_B
Seyrig	 Seyrig 021_A	 Seyrig 019_A	 Seyrig 012_A	 Seyrig 030_A
Zacos	 ZAC_0883	 ZAC_0370	 ZAC_0369	 ZAC_1575

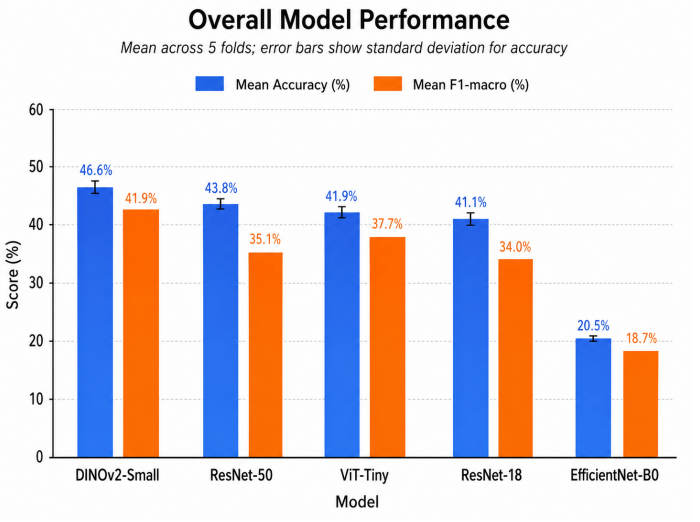
Model



Model Training Strategy

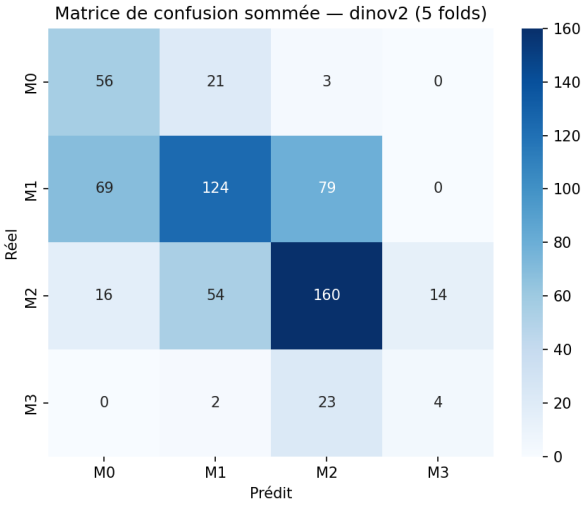
- We compared several pre-trained image models and adapted only their final layers to the seal-quality task.
- The data were evaluated with stratified 5-fold cross-validation so every fold keeps the M0–M3 distribution.
- Class weights and weighted sampling reduce the dominance of common classes.
- Data augmentations simulate photographic variation and help the model avoid memorising the small corpus.

Model Comparison



- DINOv2-Small is currently the strongest encoder.

Confusion Matrix



Discussion

- The model is strongest at identifying better-represented visual states.
- Adjacent categories, especially M1 and M2, remain difficult because they are visually close.
- M3 remains the critical weak point: very degraded seals are rare in the training data and hard even for human readers.
- These errors are informative: they show where annotation, collection expansion, and expert review are most needed.

What this Enables

- Prioritize high-quality images for automatic reading and retrieval.
- Flag low-quality or ambiguous cases for expert inspection.
- Compare collection-level quality distributions.
- Record computational uncertainty as metadata.

Limitations

- The dataset is small for deep learning: 625 images across four classes.
- Heavy class imbalance: M3 has only 29 examples.

Next steps

- Expand the corpus with new seal collections and more degraded examples.
- Synthetic generation of data.
- Test whether quality filtering improves other downstream tasks.

Conclusion

- Automatic multi-class classification of monogram quality-level across 4 seal collections.
- DINOv2-Small provides the best current baseline, but the main contribution is the quality-aware workflow.
- For digital heritage, the value is not only classification accuracy: it is making uncertainty explicit before computational interpretation begins.

Thank You!

Selected references

Digital heritage and sigillography

- Wilkinson, M. D. et al. (2016). *The FAIR Guiding Principles for scientific data management and stewardship*. Scientific Data, 3, 160018. doi:10.1038/sdata.2016.18.
- Denard, H. (ed.) (2009). *The London Charter for the Computer-Based Visualisation of Cultural Heritage*. londoncharter.org.
- Nesbitt, J. W. and Oikonomides, N. (eds.). *Catalogue of Byzantine Seals at Dumbarton Oaks and in the Fogg Museum of Art*. Dumbarton Oaks.
- BHAI Project. *Hybrid Artificial Intelligence applied to Byzantine Sigillography*, ANR-21-CE38-0001.

Computer vision methods

- He, K., Zhang, X., Ren, S. and Sun, J. (2016). *Deep Residual Learning for Image Recognition*. CVPR. doi:10.1109/CVPR.2016.90.
- Tan, M. and Le, Q. V. (2019). *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. ICML. arXiv:1905.11946.
- Dosovitskiy, A. et al. (2020). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. arXiv:2010.11929.
- Oquab, M. et al. (2023). *DINOv2: Learning Robust Visual Features without Supervision*. arXiv:2304.07193.